

Cute Goldbach Gaps

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Abstract

For an even integer N , let $\mathfrak{g}(N)$ be the least prime appearing in a Goldbach partition of N —the quantity $p(n)$ of the *minimal Goldbach partition* studied by Granville, van de Lune and te Riele, who conjectured $p(n) = O((\log n)^2 \log \log n)$ and for which $p(n) \leq 9781$ holds empirically for all even $n \leq 4 \cdot 10^{18}$. We ask how large $\mathfrak{g}(N)$ can be *forced*, and at what cost in the size of N . After an elementary warm-up showing that the neighbourhood of a factorial inflates $\mathfrak{g}$ via divisibility and Wilson’s theorem (for $N = 27! + 28$ we get $\mathfrak{g}(N) = 71$), we prove that $\mathfrak{g}$ is *unbounded*, and we do so unconditionally: combining a primorial obstruction with Dirichlet’s theorem produces, for every prime P , an even N with an *explicit* Goldbach partition and $\mathfrak{g}(N) = \text{nextprime}(P)$. We then show that a covering system of congruences forces $\mathfrak{g}(N)$ above any target Q using a modulus exponentially smaller than the primorial, and that any finite covering can likewise be realised unconditionally by a double application of Dirichlet. We exhibit a 1020-digit even integer whose least Goldbach summand is $\mathfrak{g}(N) = 100\,747$, its prime complement proved prime by an Atkin–Morain elliptic-curve certificate, and we certify a 327\,455-digit even integer for which no prime below 10^8 is a summand. To our knowledge these are, respectively, the largest *proven* least Goldbach summand on record (about ten times the previous record of 9781 near $3.3 \cdot 10^{18}$) and the largest explicit Goldbach-summand obstruction yet reported. Finally we record the *one-directional* transfer from large prime gaps: a long prime gap forces a large least summand, but the converse fails, since only prime offsets need to be covered—a sparser problem. The verified integer, the congruence system that defines it, and a standalone verification script with SHA-256 hashes are given in the appendices.

1 Introduction

Goldbach’s conjecture asserts that every even integer $N > 2$ is a sum of two primes; it has been verified for all $N \leq 4 \cdot 10^{18}$ [1]. Heuristically one expects not just one representation but many. The Hardy–Littlewood circle method—which estimates such counts by treating the primes as a random set of the appropriate density and then correcting for divisibility constraints—predicts that the number $r(N)$ of ways to write $N = p + q$ with p, q prime is

$$r(N) \sim 2 \Pi_2 \frac{N}{(\log N)^2} \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad \Pi_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right). \quad (1)$$

Reading this from left to right: $N/(\log N)^2$ is the naive guess, since a number near N is prime with probability about $1/\log N$ (the prime number theorem) and there are N choices for p ; the constant $\Pi_2 = \prod_{p>2} (1 - \frac{1}{(p-1)^2}) \approx 0.6601$ —an infinite product running over all odd primes p —is the standard correction (the twin-prime constant) that accounts for primes not being truly random mod small numbers; and the last factor $\prod_{p|N, p>2} \frac{p-1}{p-2}$, a product over the odd primes *dividing* N , inflates the count when N is highly composite. The point for us is only the conclusion: $r(N) \rightarrow \infty$,

so large even integers have *many* Goldbach partitions, and the interesting question is not whether a representation exists but how lopsided the most lopsided one must be.

Definition 1. For an even integer $N \geq 4$ admitting a Goldbach representation, the *least Goldbach summand* is

$$\mathfrak{g}(N) = \min\{p \text{ prime} : N - p \text{ is prime}\}.$$

This is exactly the quantity $p(n)$ attached to the *minimal Goldbach partition* $N = p + q$ (with p minimal) by Granville, van de Lune and te Riele [4], who conjectured that it never grows faster than a fixed multiple of $(\log N)^2 \log \log N$:

$$\mathfrak{g}(N) = O((\log N)^2 \log \log N). \quad (2)$$

The notation $f = O(g)$ means $|f| \leq Cg$ for some constant C and all large N ; here (2) is a *conjectural* ceiling, not a theorem, and one of the things we do below is locate it between provable bounds. Empirically $\mathfrak{g}$ is tiny and grows very slowly: its record values are catalogued in the OEIS [5] (sequence A020481, with records A025018 and A025019). Exhaustive verification has produced a slowly climbing ladder of records—among them $\mathfrak{g} = 1093$ at $N = 60,119,912$, then $\mathfrak{g} = 5569$ for all even $N \leq 4 \cdot 10^{14}$ [2], and the current record $\mathfrak{g} = 9781$, first attained at $N = 3,325,581,707,333,960,528$, with $\mathfrak{g}(N) \leq 9781$ for *all* even $N \leq 4 \cdot 10^{18}$ [1]. (We reconfirmed the first and last of these deterministically, both being below 2^{64} .) Strikingly, these records occur at integers of at most 19 digits. Our question is the opposite, extremal and constructive:

How large can $\mathfrak{g}(N)$ be forced to be, and at what cost in the size of N ?

A companion quantity is $S(q)$, the least even N with $\mathfrak{g}(N) = q$ [4]: the “smallest Goldbach desert” for a prescribed least summand.

Results and roadmap. We give three constructions of increasing power, and are careful throughout to separate three distinct claims: (a) no small prime is a summand of N ; (b) N has a Goldbach representation at all; and (c) the least summand equals a stated value. A covering argument gives (a) unconditionally; (b) requires either Goldbach’s conjecture, an exhibited prime complement, or a separate existence argument; and (c) requires both.

§2 **Factorial neighbourhoods.** Near $n!$, divisibility kills $N - q$ for small q , and Wilson’s theorem adds one obstruction; for $N = 27! + 28$ every prime $q \leq 67$ fails, so $\mathfrak{g}(N) = 71$.

§3 **Primorial deserts, made unconditional.** A multiple of $\prod_{p \leq P} p$ has no prime summand $\leq P$ (Lemma 6). Combining this with Dirichlet (Theorem 7) yields, for every prime P , an even N with an *explicit* partition and $\mathfrak{g}(N) = \text{nextprime}(P)$; hence $\sup_N \mathfrak{g}(N) = \infty$ unconditionally, with $\mathfrak{g}(N) \gg \log N$ by Linnik.

§4 **Covering systems.** A greedy covering of the odd primes below Q forces “no prime $< Q$ is a summand” with modulus $M \ll \prod_{p < Q} p$ (Proposition 10); any covering can be realised as an actual N with an explicit partition (Theorem 12), so $\mathfrak{g}(N) \geq Q$ unconditionally. The size advantage over the primorial *widens* with Q (Table 2).

§4.3 **Transfer from prime gaps.** A long prime gap forces a large least summand; the converse fails. The resulting lower bound is $\mathfrak{g}(N) \gg \frac{\log N \log_2 N \log_4 N}{\log_3 N}$, which together with (2) brackets the true maximal order.

We exhibit a verified 1020-digit N with $\mathfrak{g}(N) = 100\,747$ and certify a 327 455-digit N with no prime summand below 10^8 (§5); a standalone verifier with hashes is in Appendix D.

Contribution. We report two records, to the best of our knowledge. The $\mathfrak{g}(N) = 100\,747$ example is the largest *proven* least Goldbach summand: its prime complement P is certified prime by an Atkin–Morain elliptic-curve proof (Appendix B), so the value is unconditional and about ten times the previous record of 9781. The $Q = 10^8$ row is the largest explicit Goldbach-summand *obstruction*: an integer carrying a short modular certificate that rules out every prime offset below 10^8 , four orders of magnitude beyond the reach of exhaustive search. We remain careful about status: the modular certificates of absence are unconditional; the 100 747 value is proven (deterministic for 100 747, ECPP for P); and the 10^8 figure bounds $\mathfrak{g}(N)$ from below only if N is a Goldbach number at all.

Throughout, $\log_k$ is the k -fold iterated logarithm (so $\log_2 N = \log \log N$, $\log_3 N = \log \log \log N$, and so on—each extra log grows *extremely* slowly); $\theta(x) = \sum_{p \leq x} \log p$ is the sum of $\log p$ over all primes $p \leq x$ (Chebyshev’s function); $\gamma \approx 0.5772$ is Euler’s constant; $f \gg g$ means $f \geq cg$ for some constant $c > 0$ and all large arguments (the reverse of $f = O(g)$); and “BPSW” denotes the Baillie–PSW probable-prime test [12].

2 Factorial neighbourhoods and Wilson obstructions

Lemma 2 (Factorial composites). *For $n \geq 2$ and $2 \leq k \leq n$, the number $n! + k$ is composite. For $n \geq 4$ and $2 \leq k \leq n$, the number $n! - k$ is also composite.*

Proof. In both cases $k \mid n!$, so $k \mid (n! \pm k)$. For the plus side $n! + k > k$, so $n! + k$ is a proper multiple of k . For the minus side, $n! - k = k(n!/k - 1)$ with $n!/k - 1 \geq 2$ exactly when $n! \geq 3k$, which holds for $n \geq 4$; it fails at $n = 3$, $k = 3$, where $3! - 3 = 3$ is prime. $\square$

Lemma 3 (Wilson obstructions). *Let $n \geq 2$. (A) If $n + 1$ is prime, then $(n + 1) \mid n! + 1$. (B) If $n + 2$ is prime, then $(n + 2) \mid n! - 1$.*

Recall *Wilson’s theorem*: a number $p > 1$ is prime if and only if $(p - 1)! \equiv -1 \pmod{p}$, i.e. the product $1 \cdot 2 \cdots (p - 1)$ is one less than a multiple of p . We use only the forward direction.

Proof. For (A) put $p = n + 1$; Wilson’s theorem gives $n! = (p - 1)! \equiv -1 \pmod{p}$. For (B) put $p = n + 2$; then $(p - 1)! = (n + 1)n!$ and $n + 1 \equiv -1 \pmod{p}$, so $-n! \equiv -1$, i.e. $n! \equiv 1 \pmod{p}$. $\square$

Corollary 4. *If $n! + 1 > n + 1$ is prime, then $n + 1$ is composite. In particular the factorial primes $n! + 1$ (for $n = 1, 2, 3, 11, 27, 37, 41, \dots$) sit at composite $n + 1$.*

2.1 The worked example $N = 27! + 28$

Take $N = 27! + 28$, even with 29 digits. Since $27! \equiv 0 \pmod{r}$ for every prime $r \leq 27$,

$$N - q = 27! + (28 - q) \equiv 28 - q \pmod{r} \quad (r \leq 27 \text{ prime}), \quad (3)$$

so $N - q$ is killed by a small prime r exactly when $r \mid (28 - q)$: a shifted sieve that murders the class $q \equiv 28 \pmod{r}$.

Proposition 5. *For $N = 27! + 28$ every prime q with $2 \leq q \leq 67$ has $N - q$ composite, while $N - 71$ is prime; hence $\mathfrak{g}(N) = 71$.*

Proof by explicit witnesses. Table 1 gives, for each prime $q \leq 71$, a factor of $N - q$. For $q \leq 23$ the witness divides $28 - q$ via (3). Two cases below the winner are notable: $q = 29$ is killed by 29 because $29 = 27 + 2$ is prime, so Lemma 3(B) gives $29 \mid 27! - 1 = N - 29$; and $q = 59$ survives the sieve (as $59 - 28 = 31$ is 27-rough) yet $41 \mid 27! - 31 = N - 59$. Finally $71 - 28 = 43$ is 27-rough and $27! - 43$ is prime. $\square$

Table 1: Why every small prime fails as a summand of $N = 27! + 28$.

q	$N - q$	factor	q	$N - q$	factor
2	$27! + 26$	2	31	$27! - 3$	3
3	$27! + 25$	5	37	$27! - 9$	3
5	$27! + 23$	23	41	$27! - 13$	13
7	$27! + 21$	3	43	$27! - 15$	3
11	$27! + 17$	17	47	$27! - 19$	19
13	$27! + 15$	3	53	$27! - 25$	5
17	$27! + 11$	11	59	$27! - 31$	41 (generic)
19	$27! + 9$	3	61	$27! - 33$	3
23	$27! + 5$	5	67	$27! - 39$	3
29	$27! - 1$	29 (Wilson)	71	$27! - 43$	prime

By (3) a prime q can even be *tested* only if $q - 28$ is 27-rough, so the viable candidates 29, 59, 71, 89, ... are sparse, and 71 is merely the first with a prime complement. Sharper still: $27! + 1$ is prime (Corollary 4), the natural bridge $q = 27$, but 27 is composite, so one falls through to 71. For indices that are simultaneously prime and factorial-prime indices ($n = 3, 11, 37, 41$) the bridge works and $\mathfrak{g}(n! + (n + 1)) = n$. On its own the factorial construction reaches only $\mathfrak{g} \approx 2n \approx 2 \log N / \log \log N$; the next sections do far better.

3 Primorial deserts, made unconditional

The factorial wastes each small prime on a residue class. Taking N to be a multiple of a primorial wastes nothing.

Lemma 6 (Primorial core). *Let P be prime, let $m_P = \prod_{p \leq P} p$ be the primorial—the product of every prime up to P (for instance $m_{11} = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 2310$)—and let N be a multiple of m_P with $N > 2P$. Then $N - q$ is composite for every prime $q \leq P$; i.e. no prime $\leq P$ is a Goldbach summand of N .*

Proof. For a prime $q \leq P$ we have $q \mid m_P \mid N$ (as q is one of the factors of m_P), hence $q \mid N - q$, and $N - q > P \geq q$, so $N - q$ is a proper multiple of q and therefore composite. $\square$

Lemma 6 alone gives only claim (a) of the introduction. To reach unboundedness with an *actual* Goldbach number we add Dirichlet's theorem.

Theorem 7 (Unconditional unboundedness). *For every prime P put $s = \text{nextprime}(P)$. There are infinitely many primes $t \equiv -s \pmod{m_P}$, and for any such t the even integer $N = s + t$ satisfies $m_P \mid N$, has the explicit Goldbach partition $N = s + t$, and has*

$$\mathfrak{g}(N) = \text{nextprime}(P) > P.$$

Hence $\sup_N \mathfrak{g}(N) = \infty$ unconditionally. Moreover the least such t satisfies $\log N \ll P$ by Linnik's theorem [7], so $\mathfrak{g}(N) \gg \log N$.

Proof. As $\gcd(s, m_P) = 1$ (because $s > P$), Dirichlet [6] gives infinitely many primes $t \equiv -s \pmod{m_P}$. Then $N = s + t \equiv 0 \pmod{m_P}$ and N is even. By Lemma 6 no prime $\leq P$ is a summand; the only primes below $s = \text{nextprime}(P)$ are those $\leq P$, so the least summand is at least s . But $N - s = t$ is prime, so s is a summand; hence $\mathfrak{g}(N) = s$. Linnik's theorem bounds the least such prime by $t \ll m_P^L$ for an absolute constant L , and $\log m_P = \theta(P) \sim P$, so $\log N \ll P \sim s = \mathfrak{g}(N)$. $\square$

Remark 8 (Concrete instances). For $P = 13$ one has $m_{13} = 30030$, $s = 17$, and the least prime $t \equiv -17 \pmod{30030}$ is $t = 30013$, giving $N = 30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ with partition $30030 = 17 + 30013$ and $\mathfrak{g}(N) = 17$ —all primality deterministic. A larger fully proven example is the primorial $m_{43} = 13082761331670030$ itself: its minimal partition is $m_{43} = 89 + 13082761331669941$ (both proven prime, being $< 2^{64}$), so $\mathfrak{g}(m_{43}) = 89$ unconditionally.

Remark 9 (The roughness bonus). For the bare choice $N = m_P$, a prime $q > P$ makes $N - q$ coprime to all primes $\leq P$ —a property called *P-rough*—which improves its chances of being prime, since the small primes that catch most composites are already excluded. By *Mertens' theorem* (which says $\prod_{p \leq P} (1 - 1/p) \sim e^{-\gamma} / \log P$, the product of $1 - 1/p$ over primes $p \leq P$ shrinking like $1/\log P$), this sieving effect multiplies the heuristic probability by $\prod_{p \leq P} (1 - 1/p)^{-1} \sim e^{\gamma} \log P$. The expected least surviving summand then lands near $P(1 + e^{-\gamma}) \approx 1.56P$. Either way the primorial pays a modulus $m_P \approx e^P$ (its size is $\log m_P = \theta(P) \sim P$, so m_P itself is exponentially large in P); the next section removes this waste.

4 Covering systems: large deserts cheaply

4.1 The construction

A primorial spends one prime per summand; a covering system reuses each prime on a whole residue class. The one structural constraint is parity: a sum of two odd primes is even, so N is kept even and the prime 2 kills only $q = 2$; every *odd* prime below the target must be covered by odd sieving primes.

Proposition 10 (Covering certificate). *Let $Q \geq 3$ and let $\{(r, a_r) : r \in \mathcal{R}\}$ be a system of odd primes $\mathcal{R}$ such that every odd prime $q < Q$ has $q \equiv a_r \pmod{r}$ for some $r \in \mathcal{R}$. If N is even, $N \equiv a_r \pmod{r}$ for all $r \in \mathcal{R}$, and*

$$N > Q + \max_{r \in \mathcal{R}} r,$$

then $N - q$ is composite for every prime $q < Q$; i.e. no prime $< Q$ is a Goldbach summand of N .

Proof. For $q = 2$, $N - 2$ is even and > 2 . For an odd prime $q < Q$, pick $r \in \mathcal{R}$ with $q \equiv a_r \pmod{r}$; then $r \mid N - q$, and $N - q > (Q + \max_r r) - Q \geq r$, so $r \mid N - q$ properly and $N - q$ is composite. $\square$

Remark 11. The size hypothesis only has to make every divisor $r \mid N - q$ proper, for which $N > Q + \max_r r$ suffices; the much stronger $N > Q \prod_r r$ is unnecessary. A common solution N exists modulo $M = 2 \prod_r r$ by the Chinese Remainder Theorem, and any large enough even representative qualifies.

We build the covering greedily and deterministically.

Algorithm (greedy covering of the odd primes below Q).

1. Let $T = \{\text{odd primes} < Q\}$ and require $N \equiv 0 \pmod{2}$.
2. For $r = 3, 5, 7, 11, \dots$ in increasing order, while $T \neq \emptyset$: let a_r be a residue mod r containing the most elements of T (smallest such residue on ties), require $N \equiv a_r \pmod{r}$, and delete from T every $q \equiv a_r \pmod{r}$.
3. Recover N from the congruences by the Chinese Remainder Theorem.

The resulting modulus is $M = 2 \prod_{r \in \mathcal{R}} r$ —twice the product of the sieving primes actually used—which is far smaller than the primorial $\prod_{p < Q} p$ of all primes below Q , precisely because each sieving prime r is reused to kill an entire residue class of targets rather than a single one.

4.2 Realising a covering unconditionally

Proposition 10 still only gives claim (a). The same Dirichlet idea as in §3, applied twice, upgrades any covering to an actual Goldbach number.

Theorem 12 (Unconditional realisation). *Let $Q \geq 3$ and let $\{(r, a_r) : r \in \mathcal{R}\}$ be a covering of the odd primes below Q by odd primes $\mathcal{R}$. Then there are infinitely many even integers N admitting an explicit Goldbach partition $N = s + t$ (s, t prime) with $N \equiv a_r \pmod{r}$ for all $r \in \mathcal{R}$. For every such N , no prime below Q is a summand, so $\mathfrak{g}(N)$ exists and $\mathfrak{g}(N) \geq Q$.*

Proof. For each $r \in \mathcal{R}$ choose $b_r \in \{1, \dots, r-1\} \setminus \{a_r\}$, possible since $r \geq 3$. By CRT and Dirichlet there is a prime s with $s \equiv b_r \pmod{r}$ for all r ; as $s \not\equiv a_r \pmod{r}$ for any r and every odd prime $< Q$ is covered, necessarily $s \geq Q$. Next require $t \equiv a_r - s \pmod{r}$ for all r and $t \equiv 1 \pmod{2}$. Each $a_r - s \not\equiv 0 \pmod{r}$, so the combined modulus $2 \prod_r r$ and residue c satisfy $\gcd(c, 2 \prod_r r) = 1$; Dirichlet gives infinitely many primes $t \equiv c$. For any such t , set $N = s + t$: then $N \equiv a_r \pmod{r}$ for all r , N is even, and $N = s + t$ is an explicit partition. For a prime $q < Q$: if $q = 2$ then $N - 2$ is even and composite; if q is odd then $q \equiv a_r \equiv N \pmod{r}$ for some r , and for N large enough $r \mid N - q$ properly. Thus no prime $< Q$ is a summand and $\mathfrak{g}(N) \geq Q$. $\square$

4.3 Transfer from large prime gaps

Proposition 13. *If $[N - Q, N - 2]$ contains no prime, then $\mathfrak{g}(N) > Q$.*

Proof. Every prime $q \in [2, Q]$ has $N - q \in [N - Q, N - 2]$, hence composite. $\square$

So a prime gap of length $\geq Q$ just below N forces $\mathfrak{g}(N) > Q$: *prime gaps transfer to least Goldbach summands*. The transfer is, however, strictly one-directional. Forcing $\mathfrak{g}(N) > Q$ requires $N - q$ composite only for *prime* offsets $q < Q$, not for every integer in an interval; the covering of Proposition 10 covers just the primes below Q , a far sparser target. Thus “large least Goldbach summand” is a relaxation of “large prime gap,” and one should not expect the two problems to coincide.

Writing $G(X)$ for the largest gap between consecutive primes below X , the work of Ford–Green–Konyagin–Tao and of Maynard [9, 10], sharpened in [11], gives

$$G(X) \gg \frac{\log X \log_2 X \log_4 X}{\log_3 X},$$

improving Rankin [8]. In words: there are stretches of consecutive composite integers below X of length at least a constant times $\log X \cdot \log_2 X \cdot \log_4 X / \log_3 X$. The dominant factor is $\log X$

(the average prime gap near X); the three iterated-log factors are a slowly growing extra boost, and since each $\log_k$ grows glacially this is only a little more than $\log X$. Feeding the covering that underlies these bounds into Theorem 12 (which costs only a constant factor in $\log N$, by Linnik) makes the transfer unconditional:

Theorem 14. *There are infinitely many even integers N , each with an explicit Goldbach representation, such that*

$$\mathfrak{g}(N) \gg \frac{\log N \log_2 N \log_4 N}{\log_3 N}.$$

Together with the Granville–van de Lune–te Riele conjecture (2), $\mathfrak{g}(N) = O((\log N)^2 \log \log N)$, this brackets the maximal order of $\mathfrak{g}$ between a logarithmic-type lower bound and a $(\log N)^2$ -type conjectural ceiling.

5 Computational results

Method. The greedy covering was run with a vectorised sieve and N assembled by the Chinese Remainder Theorem; the construction is deterministic, so fixing the prime ordering and the tie rule reproduces every number below. Candidate complements were screened with Baillie–PSW, which is a *probable*-prime test rather than a primality certificate [12]; the headline complement was then *proven* prime by an Atkin–Morain elliptic-curve (ECP) certificate [13] (the criterion is stated in Appendix B). Complements below 2^{64} (and the small examples of Remark 8) are settled by a deterministic test.

Certified, conditional, and proven. Three statuses appear in Table 2. *Certified* (unconditional, no primality test on N): for each Q we checked exhaustively that the covering kills every odd prime below Q , so no prime $< Q$ is a summand. *Conditional*: for $Q = 10^4$ the least surviving complement is only a BPSW probable prime, so $\mathfrak{g}(N) = 10\,159$ rests on its (unproven) primality. *Proven*: for $Q = 10^5$ the complement $P = N - 100\,747$ carries an ECP certificate (Appendix B), so $\mathfrak{g}(N) = 100\,747$ is unconditional, as is the small example $\mathfrak{g}(m_{43}) = 89$ of Remark 8.

Table 2: Forcing the least Goldbach summand. “Congruences” is $|\mathcal{R}| + 1$, including the parity congruence $2:0$; “Primorial digits” is $\log_{10} \prod_{p \leq \text{gap}} p$, the size the primorial of Lemma 6 would need for the same gap; “Killed/ r ” is the average number of odd primes removed per odd sieving prime; “Shrink” is the ratio of the two size columns. [†]Conditional on the BPSW probable-primality of the complement $N - \mathfrak{g}$; [‡]proven, the complement carrying an ECP certificate. For $Q \geq 10^6$ only the certified statement (no prime $< Q$ is a summand) is claimed.

Q	$\mathfrak{g}(N)$	Congr. $ \mathcal{R} +1$	Largest r	Digits N	Primorial digits	Killed/ r	Shrink
10^4	10,159 [†]	86	443	182	4,370	14.4	24.0×
10^5	100,747 [‡]	356	2,393	1,020	43,603	27.0	42.7×
10^6	no prime $< 10^6$	1,762	15,083	6,480	433,636	44.6	66.9×
10^7	no prime $< 10^7$	9,853	102,983	44,599	4,340,851	67.5	97.3×
10^8	no prime $< 10^8$	60,627	755,137	327,455	43,424,120	95.0	132.6×

The exhibited number. The row $Q = 10^5$ gives a 1020-digit even integer N (Appendix A) for which $N - q$ is composite for all 9592 primes $q < 10^5$. Its least surviving prime is 100 747, with $P = N - 100\,747$ a 1020-digit prime, proved prime by an Atkin–Morain ECPP certificate (Appendix B). Since 100 747 is (deterministically) prime and the partition $N = 100\,747 + P$ is exhibited, $\mathfrak{g}(N) = 100\,747$ *unconditionally*. The defining congruence system is in Appendix C; the SHA-256 hashes are

N : 853e0013...e729bfc, P : 32894e4b...6e367838, ECPP cert : 9ebadbc4...41edafe,

and a standalone verifier is in Appendix D.

The certified number. The row $Q = 10^8$ defines an even integer with $\log_{10} M = 327,455$ for which $N - q$ is composite for all 5,761,455 primes $q < 10^8$, checked against the covering. This certifies, unconditionally, that no prime below 10^8 is a Goldbach summand; whether N has any summand at all is conditional on Goldbach’s conjecture. Unconditionally, Theorem 12 guarantees a (larger) even integer with an explicit partition and $\mathfrak{g} \geq 10^8$.

Reading the table. “Killed/ r ” rises from 14 to 95: each sieving prime is reused ever more heavily, the advantage a covering has over the one-for-one primorial. “Shrink” rises from $24\times$ to $133\times$: that advantage *widens* with Q . The gap of 10^8 costing the covering 327,455 digits would cost the primorial more than 43 million.

6 Discussion and open questions

We have driven the least Goldbach summand from an incidental 71 near $27!$ to a verified 100 747 and a certified 10^8 , with the unboundedness statement made unconditional through Dirichlet. The picture that emerges is a sandwich:

$$\underbrace{\frac{\log N \log_2 N \log_4 N}{\log_3 N}}_{\text{rigorous lower bound (Thm. 14)}} \ll \max_{N' \leq N} \mathfrak{g}(N') \lesssim \underbrace{(\log N)^2 \log \log N}_{\text{conjectural (GLvdtR)}}.$$

Our explicit greedy constructions sit well below even the lower-bound rate, because a greedy covering is far from the optimal Erdős–Rankin one; they are nonetheless enough to demonstrate unboundedness with concrete, certified numbers.

It is worth separating two distinct goals. One is to make $\mathfrak{g}(N)$ *large*, disregarding the size of N ; here the covering method excels and scales to obstructions (10^8) far beyond the reach of any exhaustive search. The other is the smallest-desert problem $S(q)$ —to make $\mathfrak{g}(N) = q$ with N as *small* as possible; here the method is highly suboptimal. The exhaustive record $\mathfrak{g} = 9781$ sits at a 19-digit integer, whereas our comparable $\mathfrak{g} = 10,159$ needs a 182-digit CRT integer. The value of the greedy covering is therefore not efficiency but transparency: it produces a short, independently checkable modular certificate at any prescribed scale, where exhaustive search cannot reach and where the optimal $S(q)$ remains out of analytic grasp.

1. **The smallest desert $S(q)$.** What is the least even N with $\mathfrak{g}(N) = q$ [4]? This is the natural extremal companion to the largest prime gap, and the greedy moduli here are only upper bounds on it.

2. **Removing the conditional.** The 10^5 desert is now fully unconditional, its complement ECPP-certified. The 10^8 certificate is a different matter: ruling out small summands is unconditional, but exhibiting *any* partition of that 327,455-digit integer needs a prime $q \geq 10^8$ with $N - q$ prime, then a primality proof of that $\sim 10^8$ -times-larger complement—out of reach at present.
3. **Sparser than gaps.** Does the restriction to prime offsets let $\mathfrak{g}(N)$ exceed the maximal-prime-gap rate, closing some of the distance to the conjectural ceiling (2)?

The moral is a trade-off, not a paradox: $\mathfrak{g}(N)$ can be made as large as one likes, but only by paying in the size of N , at an exchange rate governed by the growth of gaps between primes.

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A The verified integer N (1020 digits)

Even, with $g(N) = 100\,747$ (unconditional; the complement P is ECPP-certified, Appendix B). It is the Chinese Remainder Theorem solution of the system of Appendix C, reduced to its least representative exceeding 10^6 .

```
1069544539282386715486223835075605584373382706541914908831505623262975
2083522612338734637321305998987592550408498828612570022394208940143460
0064975415094707742731050071929426768009802994798394217849911437581801
4039594735967522377970546037935637237928281016918193304430368169807038
4474941987767969933359820683681557599943056143302527468556049195060358
4280425970311500284306417878555994564603068444664362599417258644737229
0484754637092494892808114334223748793668446381738325290148115072022495
7522077467401451931244937454496416229136154184084575547920258279665074
5524835562942450136969422849093934585977269428371274331010556593344957
8955544532422815286226956359279490135931859415743367621432745233231595
6982087156900220665027823763411480472052164539053448304868021268792102
4743632536289493802245261747302618432579314092247670538805639533124414
6693007026522943169287884632030399873821024301289446160936362452780885
7247690680844939813744595253510987977660444705920124097262087259987624
2505737637074064752090480111959678911758
```

B The complement $P = N - 100\,747$ (1020 digits)

A prime, proved by an Atkin–Morain elliptic-curve certificate [13] (PARI `primecert`; SHA-256 `9ebadb4...41edafe`). The certificate is a chain of steps, each invoking the following criterion.

Elliptic-curve primality criterion (Goldwasser–Kilian). Let $N > 1$ be coprime to 6, let $E : y^2 = x^3 + ax + b$ be an elliptic curve over $\mathbb{Z}/N\mathbb{Z}$, and let m, q be integers with q prime, $q \mid m$, and $q > (N^{1/4} + 1)^2$. If there is a point $\mathbf{P}$ on E with $m\mathbf{P} = O$ (the point at infinity) while $(m/q)\mathbf{P} \neq O$, then N is prime.

Here m is the number of points on E and q is a large prime factor of it; the bound $q > (N^{1/4} + 1)^2$ is what makes the implication go through. Each step thus reduces the primality of N to that of the strictly smaller prime q , which becomes the N of the next step (the “downrun”); after 120 steps the chain falls from 1020 to 16 digits, where the final q is small enough to prove directly. With 100 747 this gives the partition $N = 100\,747 + P$, so $g(N) = 100\,747$ unconditionally.

```
1069544539282386715486223835075605584373382706541914908831505623262975
2083522612338734637321305998987592550408498828612570022394208940143460
0064975415094707742731050071929426768009802994798394217849911437581801
4039594735967522377970546037935637237928281016918193304430368169807038
4474941987767969933359820683681557599943056143302527468556049195060358
4280425970311500284306417878555994564603068444664362599417258644737229
0484754637092494892808114334223748793668446381738325290148115072022495
7522077467401451931244937454496416229136154184084575547920258279665074
5524835562942450136969422849093934585977269428371274331010556593344957
8955544532422815286226956359279490135931859415743367621432745233231595
6982087156900220665027823763411480472052164539053448304868021268792102
4743632536289493802245261747302618432579314092247670538805639533124414
```

6693007026522943169287884632030399873821024301289446160936362452780885
7247690680844939813744595253510987977660444705920124097262087259987624
2505737637074064752090480111959678811011

C The covering congruence system

Each entry $r:a$ means $N \equiv a \pmod{r}$. The first, $2:0$, fixes parity; the remaining 355 odd primes (so $|\mathcal{R}| = 355$, and 356 congruences in all) cover every odd prime below 10^5 . Applying the Chinese Remainder Theorem and taking the least representative above 10^6 reproduces Appendix A.

2:0	3:2	5:3	7:5	11:8	13:6	17:4	19:11
23:10	29:7	31:17	37:20	41:19	43:12	47:37	53:35
59:37	61:22	67:11	71:6	73:60	79:43	83:6	89:59
97:7	101:75	103:83	107:33	109:87	113:24	127:53	131:57
137:2	139:31	149:51	151:61	157:53	163:31	167:124	173:33
179:40	181:162	191:62	193:109	197:51	199:33	211:30	223:209
227:18	229:32	233:123	239:152	241:3	251:42	257:185	263:139
269:18	271:21	277:211	281:242	283:235	293:101	307:30	311:54
313:150	317:15	331:199	337:145	347:250	349:125	353:29	359:47
367:101	373:288	379:82	383:338	389:319	397:83	401:106	409:318
419:64	421:143	431:15	433:124	439:3	443:27	449:231	457:249
461:41	463:142	467:82	479:3	487:104	491:33	499:243	503:51
509:165	521:214	523:496	541:39	547:201	557:518	563:548	569:446
571:127	577:499	587:45	593:20	599:463	601:78	607:402	613:305
617:182	619:16	631:42	641:43	643:302	647:7	653:283	659:479
661:60	673:198	677:67	683:500	691:155	701:46	709:42	719:152
727:334	733:284	739:282	743:257	751:70	757:364	761:541	769:542
773:36	787:28	797:182	809:7	811:704	821:506	823:802	827:86
829:580	839:4	853:173	857:91	859:398	863:287	877:216	881:14
883:112	887:123	907:283	911:8	919:235	929:249	937:585	941:96
947:10	953:347	967:950	971:68	977:551	983:179	991:638	997:71
1009:482	1013:2	1019:113	1021:880	1031:592	1033:302	1039:352	1049:803
1051:495	1061:865	1063:359	1069:216	1087:851	1091:199	1093:602	1097:748
1103:926	1109:305	1117:692	1123:457	1129:13	1151:489	1153:694	1163:34
1171:347	1181:61	1187:362	1193:79	1201:113	1213:660	1217:1121	1223:85
1229:331	1231:118	1237:779	1249:21	1259:402	1277:944	1279:47	1283:1138
1289:761	1291:292	1297:34	1301:625	1303:1076	1307:843	1319:116	1321:370
1327:209	1361:18	1367:63	1373:333	1381:555	1399:77	1409:892	1423:112
1427:82	1429:19	1433:996	1439:27	1447:862	1451:242	1453:439	1459:217
1471:148	1481:123	1483:243	1487:709	1489:175	1493:4	1499:848	1511:308
1523:267	1531:479	1543:20	1549:43	1553:445	1559:20	1567:603	1571:1
1579:89	1583:1466	1597:1133	1601:1272	1607:7	1609:163	1613:1117	1619:1100
1621:802	1627:547	1637:250	1657:702	1663:142	1667:1045	1669:55	1693:396
1697:376	1699:578	1709:1	1721:51	1723:82	1733:248	1741:196	1747:1239
1753:432	1759:708	1777:5	1783:106	1787:658	1789:1231	1801:132	1811:259
1823:548	1831:992	1847:49	1861:306	1867:71	1871:881	1873:502	1877:250
1879:1878	1889:249	1901:76	1907:552	1913:1236	1931:20	1933:1826	1949:1212
1951:525	1973:34	1979:1865	1987:586	1993:1	1997:128	1999:6	2003:1268
2011:1621	2017:5	2027:1176	2029:80	2039:1150	2053:18	2063:2	2069:997
2081:26	2083:19	2087:77	2089:20	2099:1647	2111:1541	2113:1	2129:65
2131:116	2137:73	2141:74	2143:58	2153:38	2161:2123	2179:228	2203:73
2207:739	2213:133	2221:30	2237:41	2239:9	2243:308	2251:107	2267:118
2269:176	2273:237	2281:82	2287:91	2293:112	2297:83	2309:95	2311:24
2333:606	2339:588	2341:560	2347:542	2351:1599	2357:506	2371:289	2377:457
2381:931	2383:1186	2389:1275	2393:1581				

D Standalone verification

The script below (requiring only `sympy`) rebuilds N from the system of Appendix C, checks the published SHA-256 hashes, confirms that no prime below 10^5 is a summand, and re-derives the proven example $\mathbf{g}(m_{43}) = 89$. It runs to completion with exit code 0.

```

import hashlib
from sympy import isprime, primerange
SHA_N = "853e0013...e729bfc"; SHA_P = "32894e4b...6e367838"
Q, G = 10**5, 100747
RES = [(2,0),(3,2),(5,3),(7,5),(11,8), ...] # the 356 pairs of Appendix C

def crt(pairs):
    N, M = 0, 1
    for r, a in pairs:
        N += M * (((a - N) * pow(M, -1, r)) % r); M *= r
    return N % M, M

N, M = crt(RES)
if N < 10*Q: N += M
assert hashlib.sha256(str(N).encode()).hexdigest() == SHA_N
assert N % 2 == 0
assert not [q for q in primerange(3, Q)
            if not any((N - q) % r == 0 for (r, a) in RES if r != 2)]
P = N - G
assert isprime(G) and isprime(P) # G proven; P also ECPP-certified
assert hashlib.sha256(str(P).encode()).hexdigest() == SHA_P
m43 = 1
for p in primerange(2, 44): m43 *= p
assert next(q for q in primerange(2, 1000) if isprime(m43 - q)) == 89
print("ALL CHECKS PASSED")

```

The full SHA-256 hashes of the submission artifacts (the decimal strings for N and P , and the file bytes for the congruence system and the ECPP certificate) are:

N (decimal)	853e0013d3b7627ff5e3a4a814dbc7a346a03ba3df1005fae3dee345ae729bfc
P (decimal)	32894e4bb7c8af3c8a831dfda520fc459f97a7634e696ae8217b23536e367838
residue system	502712ce8342c9998751308190afb4024ebaa5e7edab380bc8ee3225c017d088
ECPP certificate	9ebadb452ff249fede0af18aee6d3c67c3a101df39a813700235388441edafe